

3(a)	(gene L has) four alleles / L^V , L^T , L^R and L^r ; A a gene which has more than two alleles	1
3(b)	parental genotypes: $L^T L^R$ x $L^T L^R$ gametes: L^T L^R ; L^T L^R ; offspring genotypes: $L^T L^T$ $L^T L^R$ ($L^T L^R$) $L^R L^R$; offspring phenotypes: dfm-a dfm-b (dfm-b) rufipennis ; linked to genotypes ratio of offspring phenotypes: 1 : 2 : 1 ; <i>accept correct phenotype descriptions</i>	4
3(c)	L^T and L^R ; L^T and $L^R > L^V > L^r$;	2
3(d)(i)	(female) $L^V L^r$ and (male) $L^r L^r$; $L^V L^r$ (and $L^V L^r$) ;	2
3(d)(ii)	there are similar numbers of male and female offspring produced (for each phenotype) or female metallic offspring are produced or if linked to X chromosome no female metallic offspring are produced ;	1
3(d)(iii)	chi-squared test ; A χ^2	1